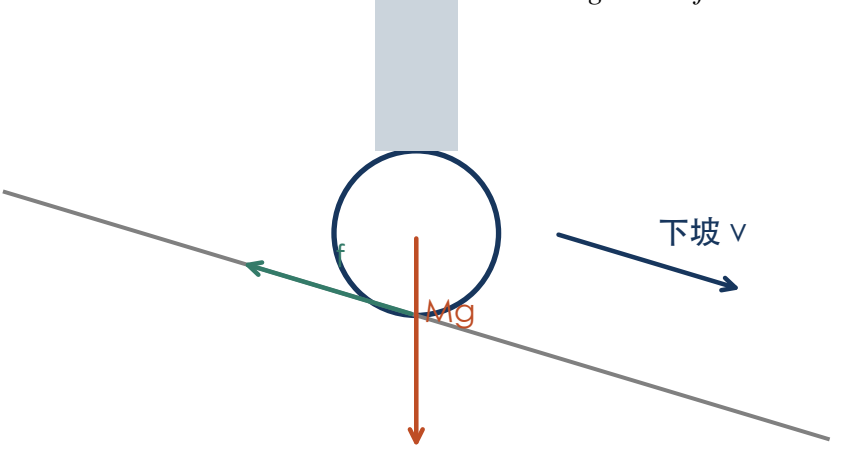
$$I\dot{\omega} = fR - \tau$$

$$Ma = Mg\sin\theta - f$$